



RAJALAKSHMI ENGINEERING
COLLEGE
An Autonomous Institution

BONAFIDE CERTIFICATE

Name: JANANI S.

Academic Year: ~~2025-2026~~ Semester:IV..... Branch: B.TECH.-IT

Register No

2116241001091

*Certified that this is the bonafide record of work done by the above student in
the SOFTWARE CONSTRUCTION [CS23432] Laboratory*

during the academic year 2025 - 2026

Signature of Faculty in-charge

Submitted for the Practical Examination held on 12/05/26

Internal Examiner

External Examiner

AZURE DEVOPS ENVIRONMENT SETUP

AIM :

To set up and access the Azure Devops environment by creating an organization through the Azure portal.

INSTALLATION :-

1. Open your web browser and go to the Azure website : <https://azure.microsoft.com/en-us/get-started/azure-portal>, sign in using your Microsoft account credentials. If you don't have a Microsoft account, you can create one here :

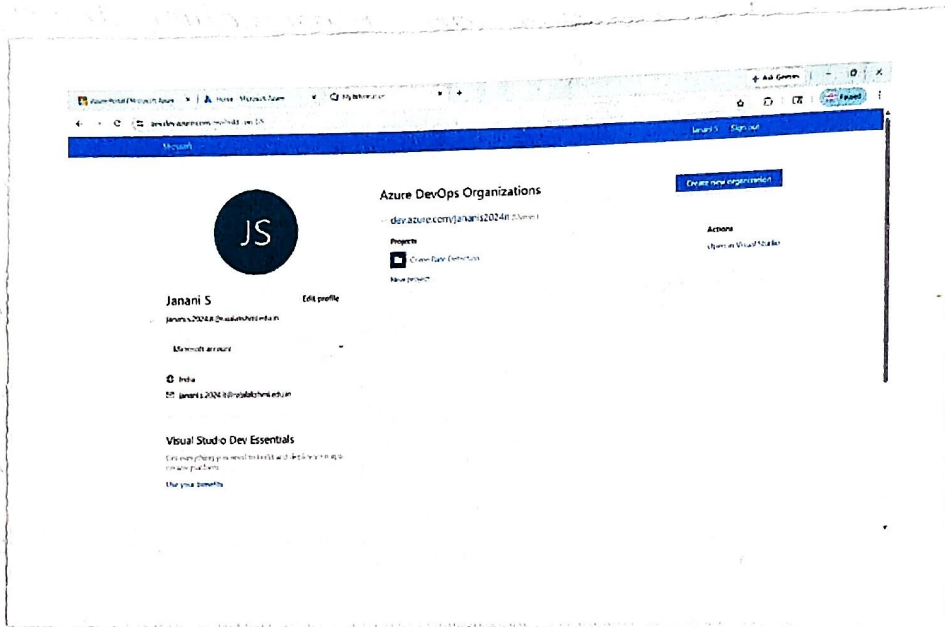
<https://signup.live.com/?lic=1>

Get started with Azure sign in

2. Azure Home Page

Azure services - Azure Devops Organizations

3. Open Devops environment in the Azure platform by typing Azure Devops organizations in the search box.



1. Get started with Azure DevOps

2. Azure DevOps

3. Azure DevOps

4. Azure DevOps

4. click on the My Azure Devops organization link and create an organization and you should be taken to the Azure Devops Organization Home page

Azure Devops

My Azure Devops Organizations

Result :-

Successfully accessed the Azure Devops environment and created a new organization through the Azure portal.

AZURE DEVOPS PROJECT SETUP AND USER STORY MANAGEMENT

AIM :-

To set up an Azure Devops project for efficient collaboration and agile work management.

PROCEDURES :

1. Create an Azure account

Azure Devops

janani.s.2024.it@rajalakshmi.edu.in

Name your Azure Devops Organization

dev.azure.com/ janani.2024.it

2. Create the first project in your Organization

(a) After the organization is set up, you'll need to create your first project. This is where you'll need to create your first project. This is where you'll begin to manage code, pipelines, work items and more.

(b) On the organizations Home page, click on the new project button.

(c) Once you have filled out the details, click create to set up your first project.

CREATE NEW PROJECT

project name : Crime Rate Detection

visibility : Public

version control : Git

work item process : janani.s.2024.it, Agile

create

Janani

janani.s.2024.it@rajalakshmi.edu.in

dev.azure.com / janani.s.2024.it

Projects - Crime Rate Detection

About this project

The Crime Rate Detection Application is a user friendly web based system design to analyze and display crime data for different locations.

Result :-

Successfully created a Crime Rate Detection project with basic data analysis and user-friendly features.

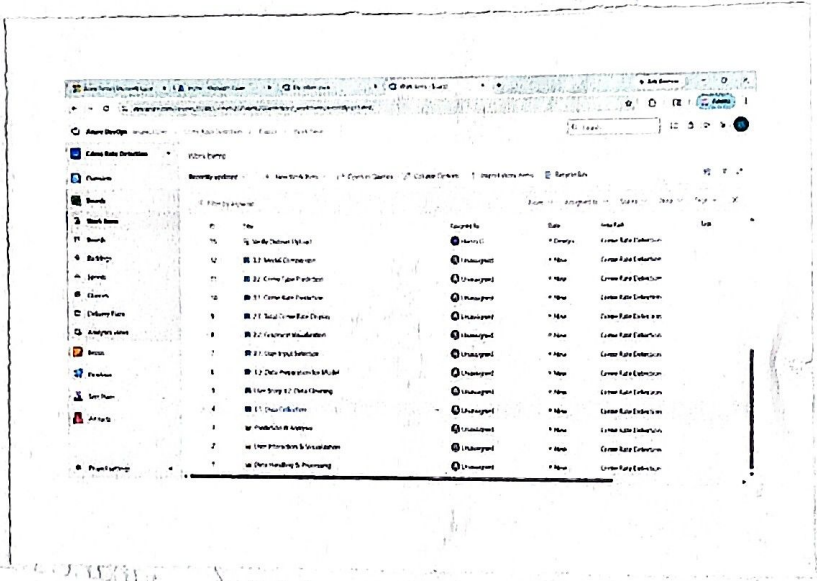
Project Name: Crime Rate Prediction

CREATE NEW PROJECT

Steps to set up your Code Project

(2) Once you have finished with the initial steps

Project Setup



Project - Crime Rate Prediction

about this project

The Crime Rate Prediction project is a user-friendly web-based system designed to analyze and predict crime rates. It is a user-friendly web-based system designed to analyze and predict crime rates. It is a user-friendly web-based system designed to analyze and predict crime rates.

Project Setup

SETTING UP, EPICS - FEATURES AND USER STORIES FOR PROJECT PLANNING

AIM:-

To learn about how to create epics, user story, features, backlogs for your assigned project.

PROCEDURE:-

create Epic, Feature, User stories, Task

As a user, I want to view crime data and trends in different locations so that I can stay aware and take necessary precautions.

1. Fill in Epic

Epic 1

Crime Data Management

2. Fill in Features

1. City/Area search Feature

2. Crime Rate Display

3. Crime Rate Analysis

4. Location-Based crime Detection

3. Fill in user story Details

- * View crime data by city/area name
- * View types of crimes in a location
- * View crime rate trends over time
- * Display user friendly interface.

User stories

* As a user, I want to search crime data by entering a city or area name so that I can know the crime details of that location.

* As a user, I want to view different types of crimes so that I can understand the nature of crimes in that area.

* As a user I want a simple and clean interface so that the application is easy to use.

Result :-

Thus, the creation of epics, features, user story and task has been created successfully.

SPRINT PLANNING

AIM:

To assign user story to specific sprint for the crime Rate Detection project.

SPRINT PLANNING:

Sprint 1:

Task

1. view crime data by city/Area

Assigned to : Janani

Iteration 1:

2. view types of crimes in a location

Assigned to : Janani

Iteration 2:

3. view crime Rate trends (daily/monthly/yearly)

Assigned to : Jayasree

4. Search location using input field

Assigned to : Jayasree

Iteration 3:

5. Detect user location automatically

Assigned to : Jaiya Nandhana

6. Handle invalid location input

Assigned to : Teiya Nandhana

Sprint 2:

7. Display crime data using charts / graphs

Assigned to : Harini

8. Show crime hotspots on map

Assigned to : Harini

Iteration 2:

9. A notification for high crime areas

Assigned to : Tanari

10. Display user friendly interface

Assigned to : Harini

Results:

✓ The sprint are successfully created for the crime rate / detection App Project.

PROJECT ESTIMATION

AIM :-

To create a proper estimation for the stories of the crime rate Detection project.

PROJECT ESTIMATION :-

Story points Measure

Effort

Complexity

Time

Using Fibonacci scale : 1, 2, 3, 5, 8, 13

Story points for Crime Rate Detection

Apps

User story	Story points	Reason
Search crime data by city	3	App Integration + input handling
View crime details of selected location	2	Simple data display
View different types of crimes	3	Data categorization

view crime

5

Data processing +

rate trends

UI charts

Deleted user

5

GPS + permission

location

handling

Handle invalid

2

Error handling

location i/p

Show user

3

UI design & styling

friendly interface

Total = 23 story points

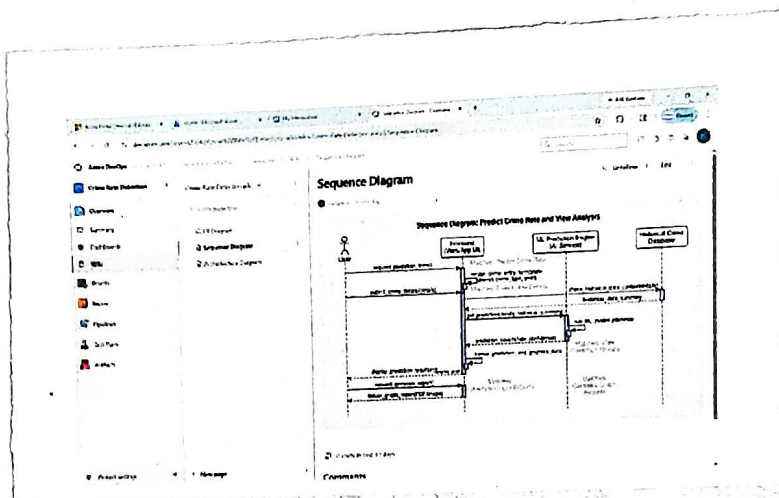
Steps :-

Boards → work items → user stories →

Assign story points → (1 / 2 / 3 / 5 / 8) → save

Result :-

The estimation is successfully created
for the crime Rate Detection project using
project estimations.



Crime rate prediction model - Crime rate prediction

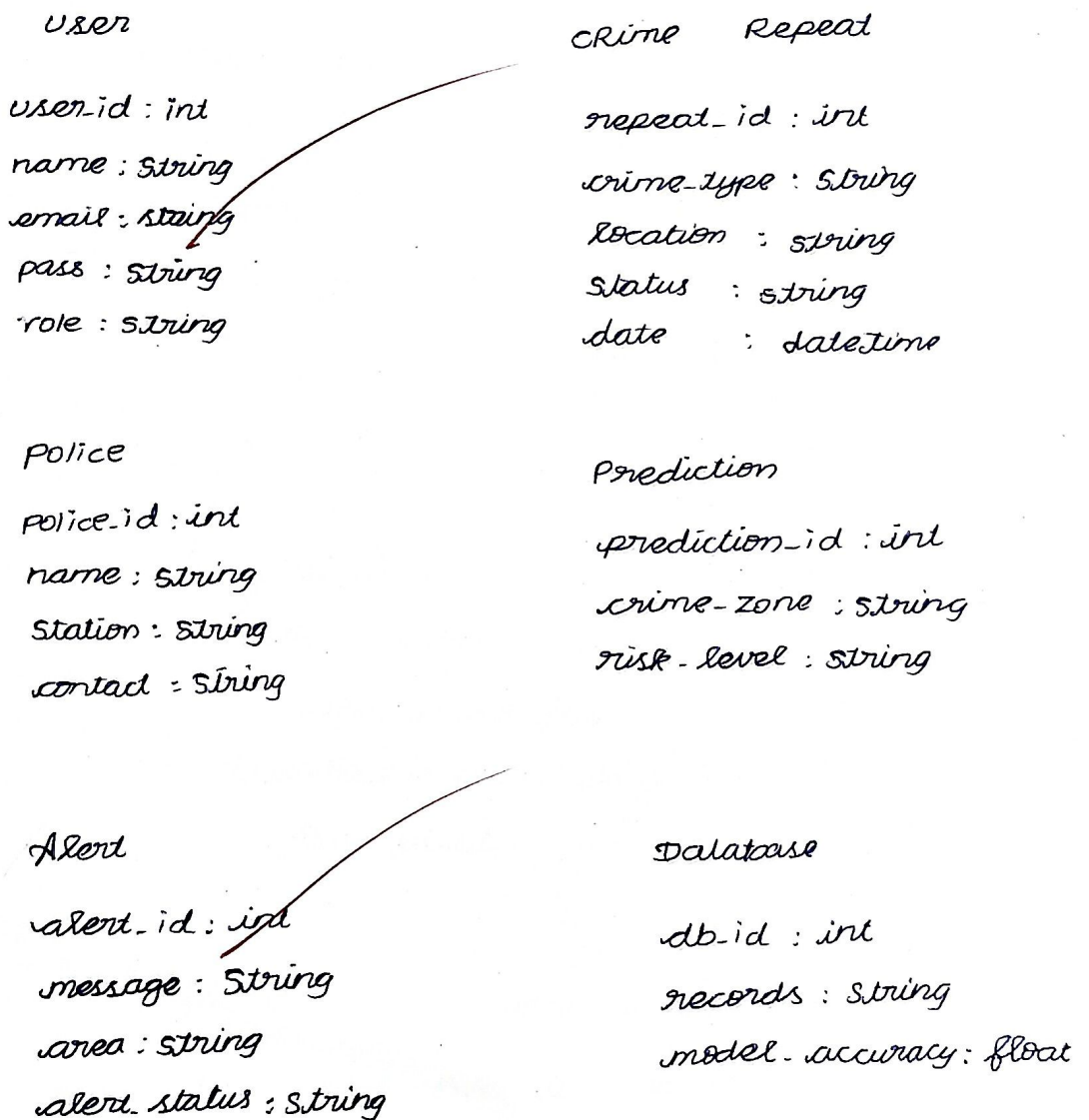
The crime rate prediction model is a machine learning model that takes in various features and predicts the crime rate for a given area.

DESIGNING CLASS & SEQUENCE DIAGRAMS
FOR PROJECT ARCHITECTURE

AIM :-

To design a class diagram and sequence diagram for the crime rate detection project.

CLASS DIAGRAM :-



SEQUENCE DIAGRAM

Actors :

User

Mobile / web App

Crime controller

Database

Flow :-

1. User → Mobile App

login (email , password)

2. Mobile App → crime controller

send login request

3. Crime controller → Database

verify user

4. Database → crime controller

Login success

5. User → Mobile App

Report crime

6. crime controller → Database

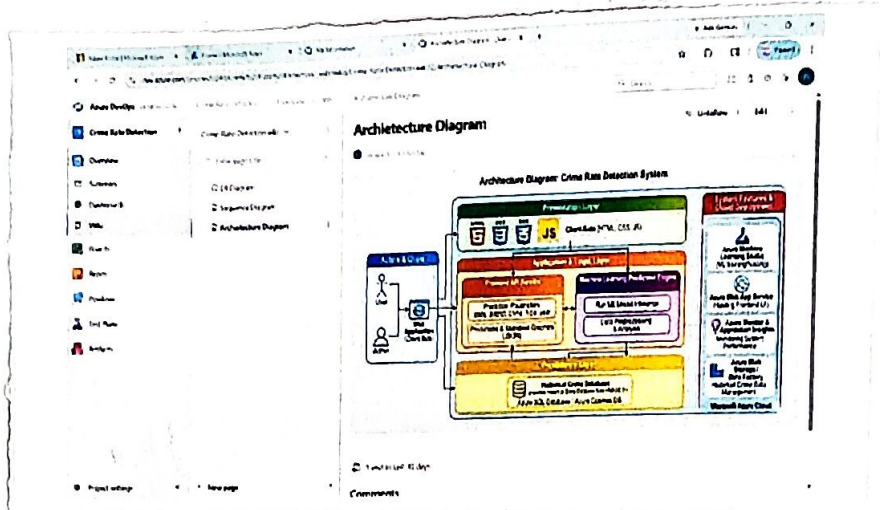
store crime data

7. Database → crime controller

Data saved

Result :-

The class and sequence diagram are designed successfully for crime rate detection.

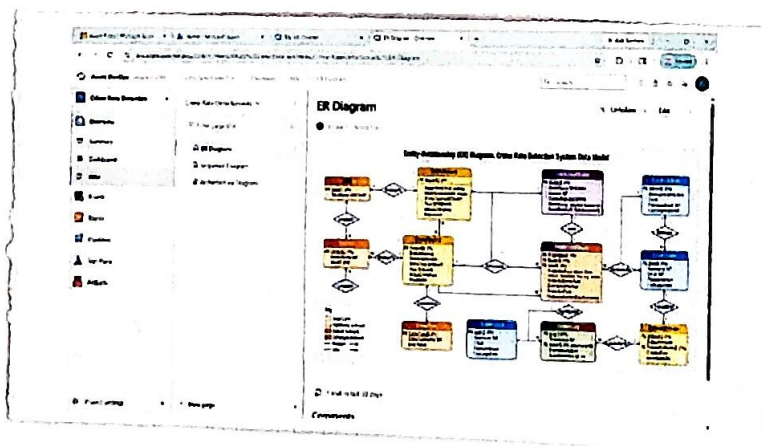


available & following services

real-time

following services & available?

real-time input



available & following services

real-time

Exp NO: 7

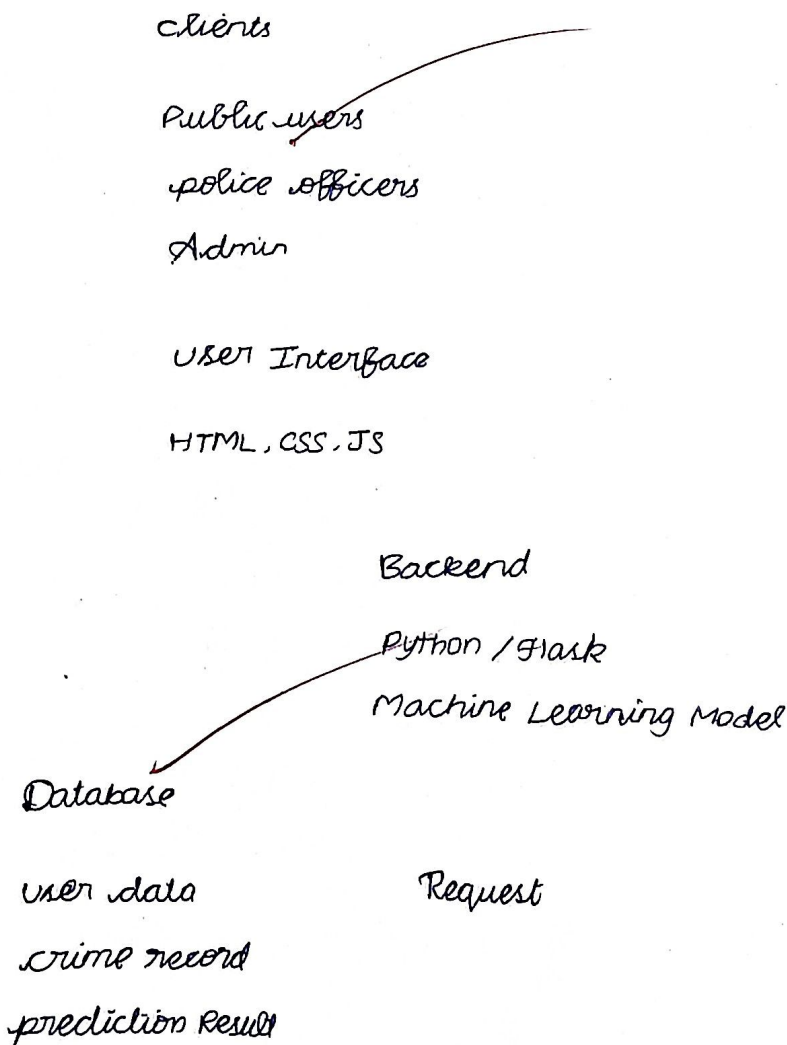
Date :- 18/03/26

DESIGNING ARCHITECTURAL & ER DIAGRAM FOR PROJECT STRUCTURE

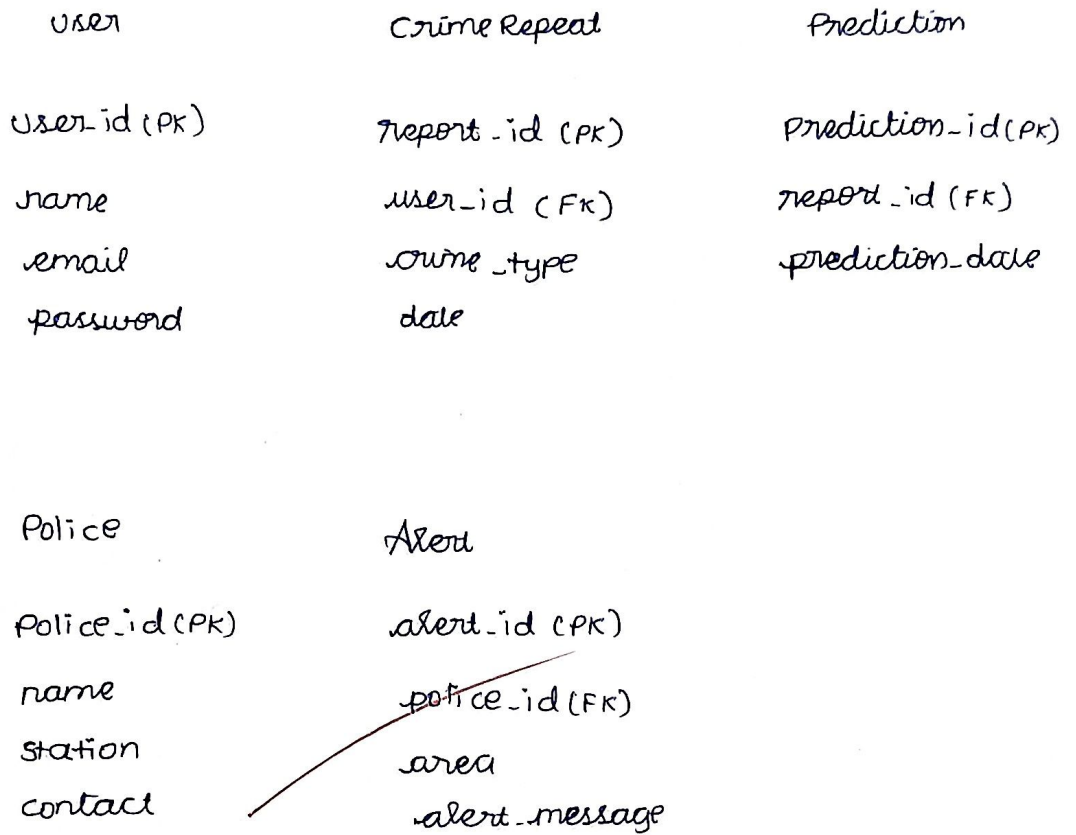
AIM :-

To design an Architectural diagram & ER diagram
for crime rate detection system.

ARCHITECTURAL DIAGRAM :-



ER DIAGRAM



RESULT :-

The architecture diagram & ER diagram is designed successfully for crime rate detection system.

TESTING - TEST PLANS & TEST CASES

AIM:-

To design test plans & test case & create test cases for at least five user stories showcasing the happy path & error scenarios in crime Rate Detection system.

TEST CODE DESIGN

1. Understand code Feature of the application

User registration & login

Report Time

Crime prediction

view crime Hotspot

2. Define User Interaction

Logging in

Reporting a crime

viewing prediction

Checking alerts

User stories

* As a user, I want to register & login, so that I can access the system.

* As a user, I want to report crimes so that authorities can take action.

* As a police officer, I want to monitor the crime

Result:-

Thus, the test plan and test cases for the crime rate detection system are created successfully.

LOADING TESTING & DETECTION

AIM:-

Aim is to perform Load Testing & implement for the crime rate detection.

LOAD TESTING:-

Test system perform under multiple users & identifies delays or issues.

Steps to create Azure load testing

- * sign in to Azure portal
- * search & click create (Azure load)
- * Fill : subscription, resource name, region
- * continue tags & click review + create
- * After deployment go to resource.

Pipelines:-

Steps :-

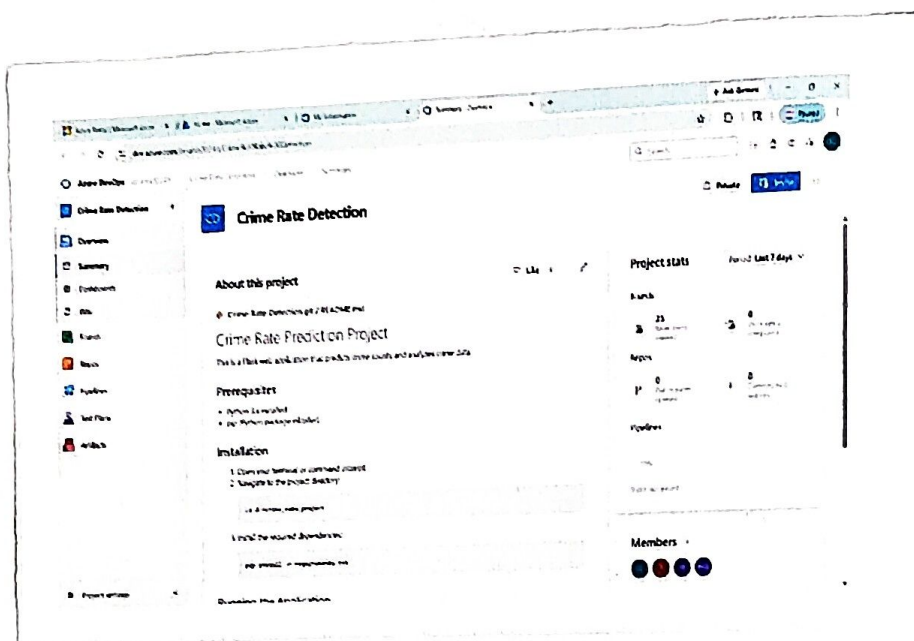
1. Conned Github repository to Azure devops
2. Create a new pipeline & select github or a service
3. Create azure-pipeline.yml in repository.

Run & Monitor pipeline

- * Commit changes to trigger pipeline
- * view page in Azure DevOps
- * Check for successful execution or errors.

RESULT :-

Successfully performed load testing & demonstrated pipeline execution for the crime rate detection project.



Exp. No: 10

Date :- 16/04/26

PROJECT STRUCTURE & NAMING CONVENTIONS

AIM:

To provide a clear & organized view of the project folder structure & file naming convention, helping contributors & users easily understand, navigate & extend the crime rate detection project. It ensures proper people management of datasets, models, output file.

Github project

<https://github.com/jananiS-Cloud/crime-rate-detection>

Result :-

The Github repo clearly displays the organized project structure & consistent naming conventions, making it easy for users & contributors to understand, maintain & navigate the crime rate detection.